

LLM Assisted SecOps: A Cognitive Assistant Framework for Cybersecurity Analysts

Harsh Dineshbhai Bakori

REVA Academy for Corporate Excellenc
REVA University
Bengaluru, India
harsh.bakori@race.reva.edu.in

Nishanth Kumar Pathi

REVA Academy for Corporate Excellenc
REVA University
Bengaluru, India
Nishanthkumarpathi@race.reva.edu.in

Shinu Abhi

REVA Academy for Corporate Excellenc
REVA University
Bengaluru, India
shinuabhi@race.reva.edu.in

Abstract—The complexity of cyber threats is rapidly outpacing the capacity of human analysts, creating an urgent need for intelligent augmentation in cybersecurity. As cybercriminals employ increasingly sophisticated tactics, traditional methods of threat detection and response are becoming inadequate. This situation necessitates a paradigm shift in how cybersecurity is approached, particularly through the integration of Artificial Intelligence (AI). Large Language Models (LLMs) can be positioned not as autonomous systems but as cognitive assistants that enhance human analysts' capabilities throughout the cyber kill chain. By translating vast, unstructured data into actionable intelligence, LLMs can provide contextual understanding, enabling analysts to make faster, more informed decisions and transition from a reactive to a proactive defensive strategy. The primary challenge lies in the overwhelming volume of data and the speed at which cyber threats evolve. The objective of this project was to explore the role of LLMs as cognitive assistants in cybersecurity, focusing on their ability to enhance the efficiency and effectiveness of human analysts. By leveraging LLMs, we aimed to improve threat identification, incident response, and overall cybersecurity posture, ultimately shifting the focus from merely reacting to threats to anticipating and mitigating them proactively. To achieve these objectives, we employed a comprehensive methodology that involved the collection of diverse datasets, including open-source intelligence, threat reports, and historical incident data. The LLM was connected to perform cybersecurity tasks using an MCP server, enabling it to generate insights and recommendations tailored to the needs of human analysts. The results of our project indicate that the integration of LLMs significantly enhances the capabilities of cybersecurity analysts. By providing real-time analysis and contextual understanding, LLMs enable faster triage of alerts and more accurate threat assessments. Our recommendations include the continued development and integration of LLMs into cybersecurity workflows, emphasizing the importance of human oversight and collaboration.

Index Terms—Artificial Intelligence, MCP server, Large Language Models (LLM), Cybersecurity, Cyber Kill Chain, Threat Intelligence, Incident Response

I. INTRODUCTION

The digital landscape is increasingly defined by a relentless battle between cybersecurity defenders and malicious actors. As cyber threats grow in volume, speed, and complexity, they outpace the capabilities of human analysts, creating significant vulnerabilities for organizations. Even the most skilled security teams are overwhelmed by alerts from numerous disparate tools, each generating a torrent of intricate data. This traditional human-centered approach to cybersecurity operations,

reliant on manual correlation and analysis, is proving inadequate against modern attacks. The resulting cognitive overload makes it difficult for analysts to monitor data, analyze emerging attack vectors, and anticipate evolving threats, leading to overlooked indicators and delayed threat identification [1], [2].

To tackle these challenges, this research aimed to redefine the role of Artificial Intelligence (AI) in cybersecurity [3]. Rather than advocating for complete automation, this project proposed a more integrated collaboration between human analysts and AI technologies. Specifically, we focus on the potential of Large Language Models (LLMs) as cognitive aids for cybersecurity professionals [4], [5]. In this framework, the LLM acts as a powerful force multiplier, designed to alleviate the cognitive burden on analysts and enhance their capabilities without replacing human intuition and expertise.

Integrating LLMs into cybersecurity workflows can significantly improve the effectiveness of cyber defense strategies [6]. By transforming vast amounts of unstructured data into actionable insights, providing real-time contextual awareness using MCP connection and tools of threats, and streamlining complex analytical processes, an LLM assistant [7] can empower analysts to make quicker and more informed decisions. This research explored how human-AI collaboration facilitated better decision-making at each stage of the cyber kill chain, ultimately leading to a more robust defense against cyber threats. The objective of this study was to present a viable approach for evolving the defensive framework from a reactive stance to a proactive, human-directed, and AI-enhanced strategy.

II. LITERATURE REVIEW

The application of Artificial Intelligence in cybersecurity is a well-established concept, with Machine Learning (ML) models playing a crucial role in tasks such as spam filtering, malware detection, and network intrusion detection. Foundational research has demonstrated the effectiveness of earlier Natural Language Processing (NLP) techniques for automating Cyber Threat Intelligence (CTI). These models have successfully parsed structured reports to extract specific indicators of compromise (IoCs), such as malicious IP addresses or file hashes, illustrating that automation can accelerate various aspects of the CTI lifecycle [8], [9]. However, while effective at pattern matching in structured data, these models struggle

with the ambiguity, context, and sheer volume of unstructured text that characterizes the modern threat landscape.

The advent of Large Language Models (LLMs) marks a significant paradigm shift in this domain. Comprehensive surveys indicate that LLMs extend beyond simple pattern recognition to achieve a deeper understanding and generation of human-like text [10]. This capability enables LLMs to process and interpret unstructured data that was previously the domain of human analysts.

Recent studies have begun to conceptualize LLMs as direct collaborators with security professionals. One notable example is "CyLens," an LLM-powered copilot designed to assist analysts throughout the entire threat management lifecycle [11]. Other studies emphasize the role of LLMs in enhancing organizational resilience by parsing vast quantities of log data and threat reports to predict potential attack vectors [12]. Despite the existing research, a significant gap remains. There is a lack of a holistic, integrated framework that defines how an LLM can function as a persistent, interactive cognitive assistant across the entire operational workflow of a cybersecurity analyst. This report aims to address that gap by proposing a model for this dynamic human-AI partnership.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

The primary challenge facing modern cybersecurity operations is the overwhelming data load placed upon human analysts. The volume and complexity of data generated by disparate security tools have surpassed the practical limits of manual analysis. This creates an environment where delayed detection, inefficient triage, and analyst burnout are common occurrences. The fundamental issue lies in the growing gap between the rapid evolution of cyber threats and the finite cognitive capacity of human analysts. This cognitive bottleneck increases organizational risk, as critical threats may be overlooked. The core problem is not a lack of data but rather a failure to translate that data into timely, actionable intelligence. An intelligent tool was required not to replace analysts but to serve as an assistant, enabling the current workforce to perform their tasks more efficiently.

B. Objectives

The primary objective of this study was to develop a theoretical framework for an LLM-based cognitive assistant designed to augment the capabilities of cybersecurity analysts [10]. The specific objectives were:

- 1) **Design a System Architecture:** Propose a high-level system architecture for an AI assistant that integrates with existing cybersecurity and LLM tools using a Model Context Protocol (MCP) server.
- 2) **Model Human-LLM Interaction:** Propose a high-level implementation of the key interaction phases between the human analyst and the LLM assistant, including query, data summarization, and proactive suggestions.

- 3) **Analyze Workflow Transformation:** Analyze how an LLM assistant could transform an analyst's tasks, focusing on improving the decision-making process at each stage of the cyber kill chain.

IV. METHODOLOGY AND SYSTEM DESIGN

A. Project Methodology

This study utilizes a theory-building methodology to systematically develop the proposed framework. It consists of four interconnected phases:

- 1) **Requirements Analysis:** Deconstructing tasks within the Lockheed Martin Cyber Kill Chain® framework to define the capabilities required of the intelligent assistant.
- 2) **Conceptual System Design:** Developing a high-level software architecture to demonstrate how the LLM assistant integrates with a standard cybersecurity ecosystem, including core components and data pipelines.
- 3) **Human-LLM Interaction Modeling:** Testing primary modes of interaction and establishing model communication protocols (MCP) to detail the flow from an initial analyst query to data summarization and proactive, context-aware suggestions from the AI.
- 4) **Workflow Transformation Analysis:** Analyzing the transformative impact of the proposed assistant by creating a proof of concept (POC) and evaluating its effect on an analyst's workflow and decision-making.

B. System Design

This system is built on a modular framework designed to separate distinct functions and facilitate secure, scalable operations. As illustrated in Fig. 1, the architecture's core is the central MCP Service Layer. This layer acts as an intelligent gateway, utilizing various external tools for in-depth analysis while ensuring the AI Interface has access to the latest data [13].

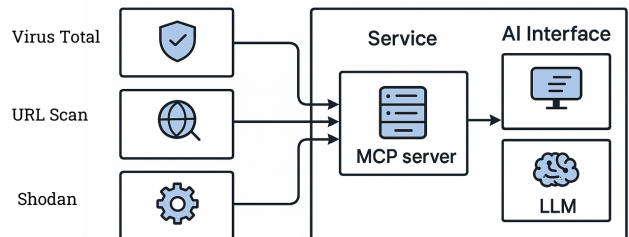


Fig. 1. High-level conceptual design of the LLM assistant system.

The system components are:

- **AI Interface:** The primary point of contact for the analyst, consisting of a User Interface (UI) and an LLM to interpret commands, formulate a strategy, and retrieve data.
- **MCP Service Layer:** The system's orchestrator, which receives requests from the AI Interface and delegates

tasks. A critical component is the MCP Server, a specialized gateway that manages access to sensitive data and tools [14].

- **External Tools:** A suite of analytical services like VirusTotal and URLScan.io. The Service Layer interacts with these tools through their APIs to provide up-to-date threat intelligence.

This design allows the LLM to intelligently utilize specialized external resources indirectly, while the Service Layer maintains current data throughout the process.

V. IMPLEMENTATION

The implementation of the LLM-based cybersecurity assistant was structured into several key phases to develop the proof of concept (POC). The hardware foundation included an NVIDIA GPU with 8GB of VRAM for efficient AI model inference. The software stack consisted of an open-source LLM (Gemini-2.0-flash), a Python-based backend using FastAPI and fastMCP, and OpenWebUI for the user interface.

The core of the implementation was the MCP Service Layer, configured as the central orchestrator to manage API requests to external tools like VirusTotal and URLScan.io. These tools were integrated as callable functions, allowing the LLM to access real-time threat intelligence. Interaction protocols were established to ensure the LLM could effectively interpret analyst commands, summarize data, and provide proactive suggestions. The final phase involved comprehensive testing to validate the system's functionality and performance.

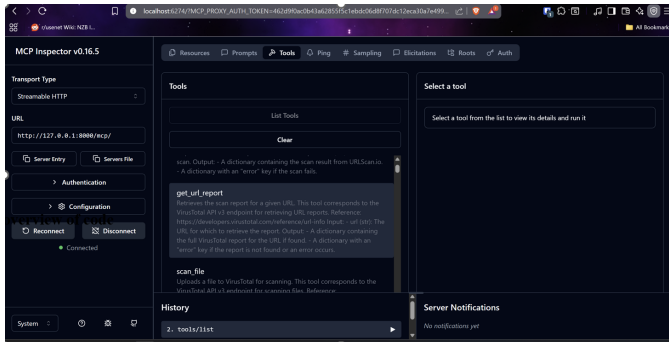


Fig. 2. MCP Inspector showing available tools connected to the server.

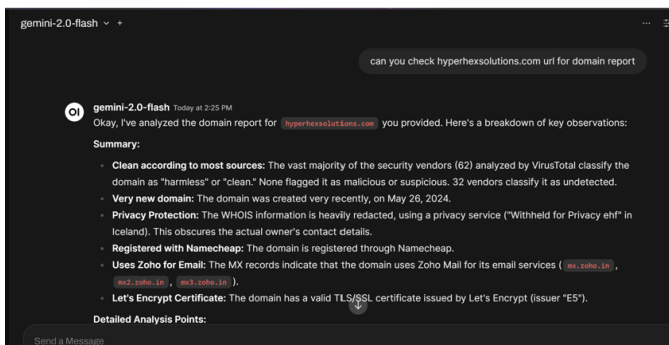


Fig. 3. LLM using connected tools to generate a domain report for an analyst.

Fig. 2 shows the MCP Inspector UI, which lists the tools made available to the LLM via the server. Fig. 3 demonstrates the final output, where the LLM has analyzed a URL using the integrated tools and provided a summarized report to the analyst.

VI. TESTING AND VALIDATION

To validate the assistant's effectiveness, we conducted structured, scenario-based tests simulating realistic cyber threats.

A. Scenario 1: Spear-Phishing Campaign

An analyst was tasked with investigating a reported phishing email containing a link to a credential harvesting page. The analyst asked the LLM assistant to check the URL for malicious intent using tools like URLScan.io and VirusTotal, and to summarize threat intelligence on the source IP using AbuseIPDB. The assistant successfully identified the URL as malicious, retrieved the scan report, and provided context on the IP's reputation, significantly speeding up the initial triage process.

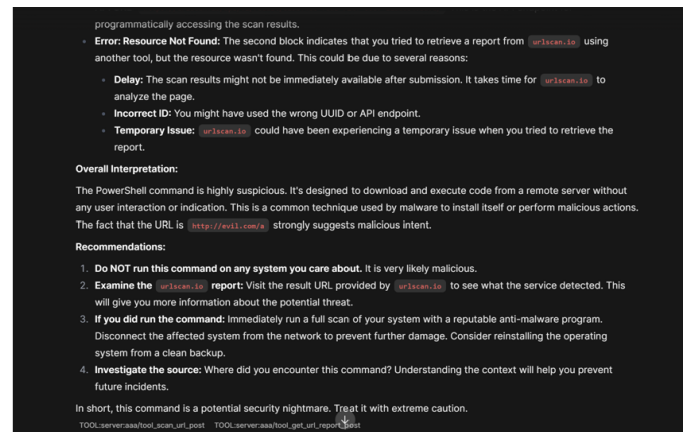


Fig. 4. Result of Test Scenario 1: URL analysis for phishing.

B. Scenario 2: Malware Infection

An EDR alert flagged a suspicious PowerShell command, and network monitoring tools flagged unusual outbound traffic. The analyst provided the command to the assistant for an explanation and checked a file hash for malware reports. The LLM correctly identified the command as a technique to download and execute code from a remote server. It provided immediate recommendations for containment and used tools like ThreatFox and AlienVault OTX to investigate the associated indicators.

As seen in Fig. 4 and Fig. 5, the assistant provided accurate, actionable intelligence in both scenarios, validating its utility in real-world investigations.

VII. ANALYSIS AND RESULTS

The evaluation confirmed both the functional performance and the viability of the proposed software architecture. By successfully integrating the LLM with external online tools

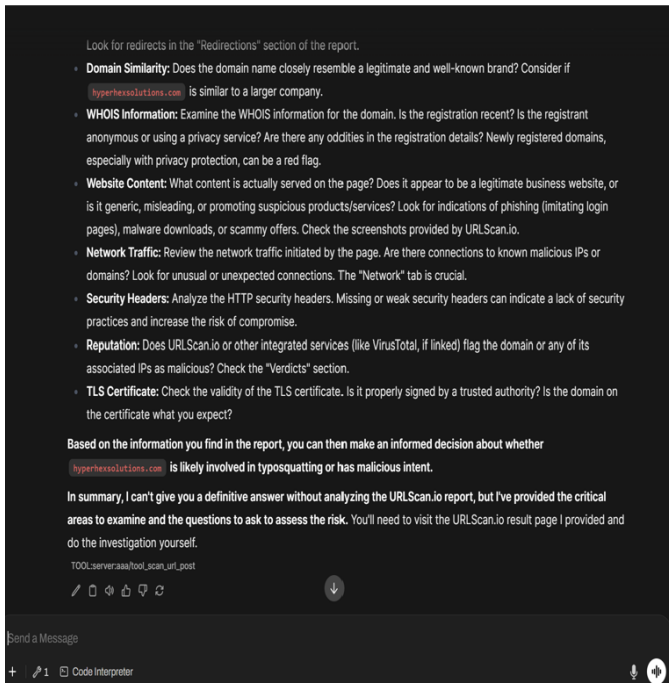


Fig. 5. Result of Test Scenario 2: PowerShell command analysis.

through the central Service Layer, the system demonstrated a significant enhancement in operational speed. Analysts were able to pose high-level questions and receive immediate, synthesized answers, enabling them to reach accurate conclusions in a fraction of the time typically required.

The human-AI partnership was shown to be highly effective. The assistant functioned as an "on-demand expert," bridging knowledge gaps and democratizing expertise. Less experienced analysts were able to use the system effectively, as the assistant provided instant context on unfamiliar threats. This model alleviated the cognitive load on analysts, shifting their role from manual data analysis to asking the right questions. By offloading the burden of data gathering, the assistant allowed analysts to transition from reactive "firefighting" to proactive threat hunting. The project successfully met its objectives, demonstrating a clear transformation in the analyst's workflow and validating the core thesis that an LLM assistant is a powerful tool for augmenting human capabilities [15].

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The implementation of the LLM-based cybersecurity assistant has shown improvements in the efficiency and effectiveness of security analysis. By integrating an LLM within a modular architecture, the system has transformed how analysts interact with data, moving from traditional manual processes to a more streamlined and intelligent approach. The results indicate that the assistant not only delivers synthesized, actionable intelligence but also reduces cognitive load, allowing analysts to concentrate on strategic decision-making. The system's ability to deliver intelligence on demand shortens the timeline

for detecting and responding to threats and empowers analysts at all experience levels. The study's main premise—that an LLM can be a tool that amplifies the skills of the human analyst—was effectively confirmed.

B. Future Scope

The successful demonstration of this framework leads to several opportunities for future research and development.

- 1) **Integration with Additional Tools:** Expanding the range of integrated tools to include threat intelligence feeds, SIEMs, and incident response platforms would further enrich the assistant's analytical capabilities.
- 2) **Advancing the Proof-of-Concept:** Maturing the current POC into a more robust prototype suitable for a production environment. This would require hardening the software and conducting larger-scale usability tests with dedicated security teams.
- 3) **Investigating Supervised Automation:** Cautiously examining the possibility of giving the assistant a degree of safe, supervised autonomy for low-risk, high-confidence tasks, such as blocking a known malicious IP after receiving one-click approval from an analyst.

By pursuing these directions, the LLM-based cybersecurity assistant can evolve into an even more powerful tool, further enhancing the resilience of cybersecurity operations.

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